

Steve Tonneau

Reader
Legged Robotics and Computer Graphics

KEYWORDS

Motion planning and synthesis, optimal control, humanoid robotics, procedural animation

FUNDINGS AND AWARDS

Principal Investigator - TCS Grant - 500k (23-25)
PI on H2020 Memmo Project (18-22):

Stars of Europe prize in Paris for exceptional
EU-funded collaborative research.

Grand Prize for ANR Entracte (13-17)
RSI Exchanges Award (2018)

RESEARCH POSITIONS

SINCE JANUARY 2019

University of Edinburgh, Scotland

Reader (associate professor)

MARCH 2015 - DECEMBER 2018

LAAS-CNRS, Toulouse, France

Post-Doctorate – national project Entracte (ANR)

DECEMBER 2011 - FEBRUARY 2015

IRISA, Rennes, France

Phd - diploma obtained February 2015

Autonomous locomotion for virtual characters in
constrained environments.

COMMUNITY SERVICE AND DUTIES

- Deputy director, D2AIR CDT - since 24
- Translation lead and CDT RAS expert - 23-24
- Post-doc champion - 24 - 26
- Year Organiser - Edinburgh - 20 - 24
- Associate editor for IROS and ICRA.
- Reviewer for all major Robotics venues

TEACHING POSITIONS

University of Edinburgh / Insa Rennes /
Supaero Toulouse

- Advanced robotics (Since 2022)
- System Design Project (2019 – 2022)
- Robotics: motion planning and control (2015 – 2018)
- Various computer science courses (2011 – 2015)

INDUSTRY CONSULTING

MARCH-MAY 21

Nimble one - Toulouse

Technology transfer - control of a legged robot

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🌐	stevetonneau.fr

ENGINEER EXPERIENCE (3.5 YEARS)

JULY 2008 – OCTOBER 2011

Masa Group, Paris, France

Software Engineer then Project Manager

EDUCATION

- | | |
|-------------|---|
| 2005 – 2008 | Master in Computer science
INSA ENGINEERING SCHOOL
Rennes, France |
| 2007 – 2008 | “Game design and Development” Master classes
ABROAD SEMESTER AT RIT
Rochester, USA |
| 2003 – 2005 | 2 year diploma on Mathematics
UNIVERSITY OF MONTPELLIER II
Montpellier, France |

COMPUTER SCIENCE SKILLS

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|----------------------|--|
| ENGINES AND SOFTWARE | Blender, Unity 3d,
Morpheme, ODE,
Bullet, Mujoco, Isaac* |
| LANGUAGES | C++, Python, Java,
C#, Prolog, Lua,
Scheme, O-Caml |
| DATA | PostgreSQL, PostGIS |

CURRENT STUDENTS

- | | |
|-------------------|-------------------------|
| VICTOR LEVÉ-LIN | PhD student |
| CIARAN JOHNSON | PhD student |
| JULIA LOPEZ GOMEZ | PhD student |
| BEN BRADWELL | Msc by research student |

PAST PHD STUDENTS AND POSTDOCS

- | | |
|----------------|-----------------|
| DANIEL LAYEGHI | THOMAS CORBERES |
| JASON CHEMIN | JYAYI WANG |
| TRAIKO DINEV | PIERRE FERNBACH |

WORKSHOPS

IROS '16

Towards a unified workflow for multi contact motion on legged robots: Challenges in planning, optimization and control

Tonneau, Bretl, Mansard

Main organizer of the workshop ([link](#)).

SELECTED JOURNAL PUBLICATIONS*

IEEE ACCESS

Perceptive locomotion through whole-body MPC and optimal region selection

Corbères, Mastalli, Merkt, Havoutis, Fallon, Mansard, Flayols, Vijayakumar, Tonneau

TRANSACTION ON ROBOTICS (T-RO)

Inverse-dynamics mpc via nullspace resolution

Mastalli, Chhatoi, Corbères, Tonneau, Vijayakumar

TRANSACTION ON ROBOTICS (T-RO)

Online Multi-Contact Receding Horizon Planning via Value Function Approximation

Wang, Kim, Santoso Lembono, Du, Shim, Samadi, Wang, Ivan, Calinon, Vijayakumar, Tonneau

T-RO

C-CROC: Continuous and Convex Resolution of Centroidal dynamic trajectories for legged robots in multi-contact scenarios

Fernbach, Tonneau, Stasse, Carpentier, Taïx

TRANSACTION ON GRAPHICS

2PAC: Two Point Attractors for Center of Mass Trajectories in Multi Contact Scenarios

Tonneau, Del Prete, Pettré, Mansard

T-RO

An efficient acyclic contact planner for multiped robots

Tonneau, Del Prete, Pettré, Park, Manocha, Mansard

T-RO

Zero Step Capturability for Legged Robots in Multi Contact

Del Prete, Tonneau, Mansard

SELECTED CONFERENCE PAPERS*

ICRA '26

CASSR: Continuous A-Star Search through Reachability for real time footstep planning

Wang, Tonneau

HUMANOIDS '24

NAS: N-step computation of All Solutions to the footstep planning problem

Wang, Samadi, Wang, Fernbach, Stasse, Vijayakumar, Tonneau

ICRA '22

Convex strategies for trajectory optimisation: application to the Polytope Traversal Problem

Tonneau

ICRA '22

Optimal Control via Combined Inference and Numerical Optimization

Layeghi, Tonneau, Mistry

IROS '21

Real-time footstep planning and control of the solo quadruped robot in 3d environments

Risbourg, Corbères, Léziart, Flayols, Mansard, Tonneau

ICRA '20

SL1M: Sparse L1-norm Minimization for contact planning on uneven terrain

Tonneau, Song, Fernbach, Mansard, Taïx, Del Prete

IROS '18

CROC: Convex Resolution Of Centroidal dynamics trajectories to provide a feasibility criterion for the multi contact planning problem

Fernbach, Tonneau, Taïx

ICRA '18

Using a Memory of Motion to Efficiently Warm-Start a Nonlinear Predictive Controller

Mansard, Del Prete, Geisert, Tonneau, Stasse

ICRA '16

A versatile and efficient pattern generator for generalized legged locomotion

Carpentier, Tonneau, Naveau, Stasse, Mansard

IROS '16

HPP: a new software for constrained motion planning

Mirabel, Tonneau, Fernbach, Seppälä, Campana, Mansard, Lamiroux

*All my publications can be found on my website (including pre-prints) or on google scholar